World Cellular Automata

A Recursive Local-World Architecture for World-State Modeling
Technical Report v0.1

WCA Research Project

2026-07-01

Abstract

World Cellular Automata (WCA) is a recursive architecture for world-state modeling. Instead of mapping an observed state directly to a future state, WCA expands a global state into a family of node-centered local worlds, evolves those local worlds through learned pair interactions, and recomposes their centers into the next global state. This technical report introduces the WCA architecture, gives its finite-set and tensor semantics, and records a staged experimental program across mazes, WeatherBench-style fields, and PDEBench reaction-diffusion. The central claim is architecture-first: recursive local-world synthesis is a viable and testable primitive for structured world-state prediction. The current evidence is strongest in token-level PDEBench reaction-diffusion, where recursion depth, learnable field interfaces, attribution controls, and h8x2 rollout diagnostics indicate that the WCA core can contribute beyond interface-only and no-recursion alternatives. The report also records negative results and unresolved boundaries, including larger-token PatchMean underperformance, native-resolution visual limits, and systems scaling constraints. These limitations are not treated as failures of the research direction, but as the next experimental frontier.

Contents

1 Purpose of This Technical Report

This document is a technical report, not a compressed conference submission. Its purpose is to make the WCA research program legible and reproducible: what the architecture is, why it is worth studying, what has been tested, what the experiments show, what they do not yet show, and which next steps are scientifically justified.

The report is intended for public release on GitHub, Hugging Face, and Zenodo. It therefore favors completeness over page economy. It includes architecture motivation, mathematical notation, shape contracts, experimental protocol, selected results, negative findings, and a roadmap. Traditional algorithm-box pseudocode is intentionally omitted. WCA is not best understood as a generic training loop. Its important property is the semantic structure of the intermediate local-world tensor and the preservation of the center index during pair evolution.

2 Why Study WCA?

Physical and embodied prediction tasks require both local interaction and global coherence. Reaction-diffusion, contact dynamics, heat transfer, shortest-path navigation, and weather fields all involve local rules whose effects accumulate into global states. A model that treats the future as a single direct output can perform well, but it often hides the mechanism of state evolution inside a black-box map. A purely local cellular automaton has a clear mechanism, but it can struggle to retain global context over long horizons.

WCA proposes a middle path. Each node is allowed to construct a local interpretation of the entire world, evolve that local world through pairwise relations, and then contribute its evolved center back to the shared world state. This gives the model a concrete recurrent structure:

$$H_t \longrightarrow L_t \longrightarrow L'_t \longrightarrow H_{t+1}.$$

The motivation is not only empirical. WCA is attractive because it makes a specific architectural wager:

> World-state prediction can be improved by repeatedly differentiating a global state into node-centered local worlds and recomposing those evolved local worlds into a shared next state.

This wager is falsifiable. If recursion depth does not matter, if no-recursion controls match the full core, if tokenizer-only controls explain the gain, or if long-horizon rollout collapses, then the WCA hypothesis weakens. The experiments in this report are organized around these tests.

3 Mathematical Formulation

3.1 World state

Let the represented world have a finite node set

$$\mathcal{V} = \{1, 2, \ldots, N\}.$$

At time t, WCA represents the world as a hidden state function

$$h_t : \mathcal{V} \to \mathbb{R}^D,$$

or, in batched tensor form,

$$H_t \in \mathbb{R}^{B\times N\times D},$$

where B is batch size, N is the number of nodes or field tokens, and D is the hidden dimension.

3.2 Local worlds

The defining move of WCA is to expand the world state into a family of local worlds:

$$L_t = \Pi_\theta(H_t), \qquad L_t \in \mathbb{R}^{B\times N_c\times N_v\times D}.$$

In the protected dense baseline, $N_c = N_v = N$. The first node axis indexes the center c of a local world. The second node axis indexes the represented world node r inside that centered world. In function notation, a local world is

$$\ell_{t,c} : \mathcal{V} \to \mathbb{R}^D,$$

and the full bundle of local worlds is

$$\{\ell_{t,c}\}_{c\in\mathcal{V}}.$$

This is the key distinction from ordinary message passing. WCA does not only update $h_t(i)$ from neighbors of i. It constructs a center-indexed family of world interpretations and evolves each interpretation before recomposition.

3.3 Pair evolution inside local worlds

Inside a local world centered at c, WCA evolves receiver node r by aggregating pair interactions with sender nodes s:

$$\ell_{t,c}^{k+1}(r) = \ell_{t,c}^{k}(r) + \sum_{s\in\mathcal{V}} A_{rs}\,\phi_\theta\left(\ell_{t,c}^{k}(r), \ell_{t,c}^{k}(s), e_{rs}\right).$$

Here A_{rs} is a support or adjacency weight, e_{rs} is optional edge information, ϕ_θ is a learned pair kernel, and k indexes inner pair-evolution steps. The full dense semantics are equivalent to an interaction over

$$(b, c, r, s, d) \in \{1, \dots, B\} \times \mathcal{V} \times \mathcal{V} \times \mathcal{V} \times \{1, \dots, D\},$$

which has semantic shape

$$[B, N, N, N, D].$$

Implementations may chunk or reorder computation for memory, but the protected baseline claim depends on preserving this semantic structure.

3.4 Center recomposition

After K inner pair-evolution steps, WCA recomposes the next world state from the evolved center diagonal:

$$h_{t+1}(i) = \rho_\theta\left(\ell_{t,i}^{K}(i)\right),$$

or, in tensor shorthand,

$$H_{t+1} = \rho_\theta\left(\operatorname{diag} L_t^K\right).$$

The center index is therefore not an implementation accident. It is a semantic carrier. The local world centered at i contributes the evolved representation of node i to the next global state.

3.5 Outer recursion

The previous definition gives one outer update. WCA can apply the update multiple times:

$$H_t^{(m+1)} = F_\theta\left(H_t^{(m)}, A, E\right), \qquad m = 0, \ldots, M-1.$$

The final prediction is read from $H_t^{(M)}$. In the experiments, M is called outer recursion depth. The V25 recursion-depth ladder tests whether larger M improves long-horizon physical prediction.

4 Shape Contract and Architecture Invariants

Object	Shape	Meaning
H_t	$[B, N, D]$	Global world state at time t.
L_t	$[B, N, N, D]$	Center-indexed local-world bundle.
Pair semantics	$[B, N, N, N, D]$	Batch, center, receiver, sender, channel.
diag L_t	$[B, N, D]$	Evolved centers used for recomposition.
H_{t+1}	$[B, N, D]$	Next global world state.

The protected WCA baseline must preserve the following invariants:

1. The model maintains a world state $H_t \in \mathbb{R}^{B \times N \times D}$.
2. The model constructs local worlds $L_t \in \mathbb{R}^{B \times N \times N \times D}$.
3. Pair evolution acts inside local worlds and preserves the center index.
4. Recomposition extracts evolved centers into H_{t+1}.
5. Memory optimization must preserve dense semantics unless declared as an explicit variant.

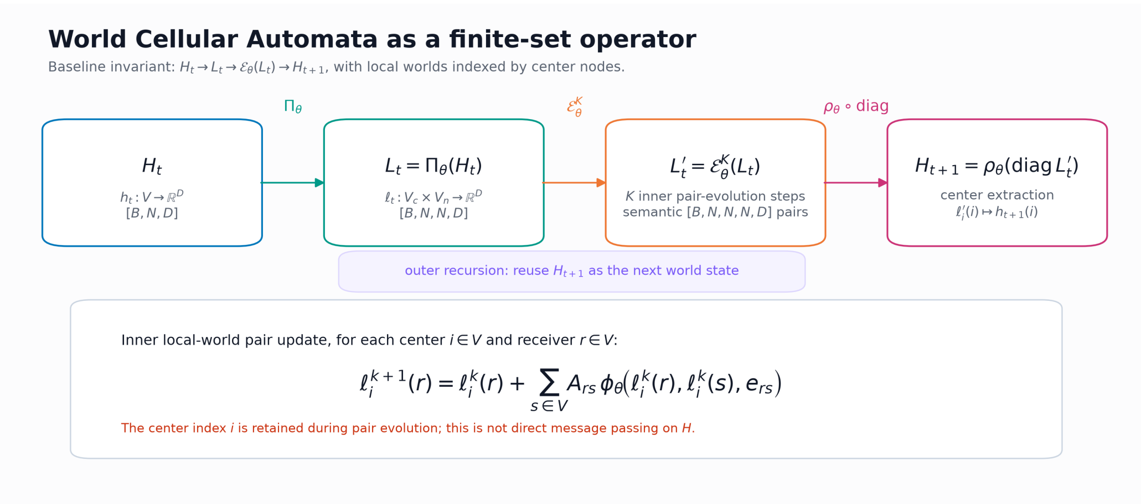

Figure 1: WCA as a finite-set operator. The protected semantic invariant is $H_t \to L_t \to \mathcal{E}_\theta(L_t) \to H_{t+1}$. The key intermediate object is the center-indexed bundle of local worlds.

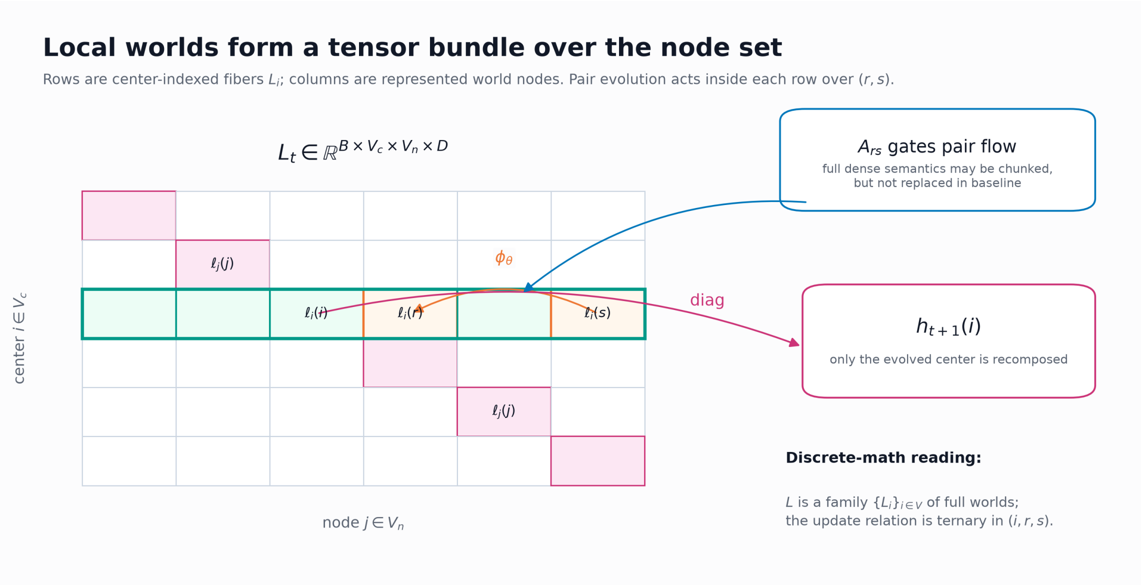

Figure 2: Local worlds form a tensor bundle over the node set. Pair evolution acts inside a row indexed by center c over receiver and sender positions (r, s); recomposition uses the evolved diagonal.

5 Relation to Other Model Families

5.1 Versus neural cellular automata

Neural Cellular Automata learn local update rules over grid states. WCA shares the intuition that repeated local transformations can produce global structure, but WCA is not only a local grid rule. Each center c carries a local world ℓ_c over the full represented node set. The update is therefore a local-world evolution, not simply a neighborhood update around one grid cell.

5.2 Versus graph neural networks

A message-passing GNN typically updates node i using messages from neighboring nodes j:

$$h_i^+ = \psi_\theta\left(h_i, \sum_{j \in \mathcal{N}(i)} \phi_\theta(h_i, h_j, e_{ij})\right).$$

WCA instead evolves $\ell_c(r)$ inside every center-indexed local world:

$$\ell_c(r)^+ = \ell_c(r) + \sum_s A_{rs}\phi_\theta(\ell_c(r), \ell_c(s), e_{rs}).$$

The additional center index c is the architectural difference. Removing it collapses WCA toward ordinary graph message passing and must be treated as an ablation, not as an equivalent implementation.

5.3 Versus transformer attention

Transformer attention computes data-dependent weighted interactions among tokens. WCA also learns relations among nodes, but it does not rely on a standard query-key-value attention block. Its distinctive structure is the creation, evolution, and recomposition of local worlds. Attention could be used inside a future pair kernel variant, but that would be a variant of WCA, not the protected baseline described here.

5.4 Versus neural operators

Neural operators learn maps between function spaces. They are strong baselines for PDE surrogate modeling because they can efficiently map present fields to future fields. WCA asks a different question: can a recursive state machine over local-world bundles learn useful world dynamics? The comparison is not "WCA replaces FNO." The comparison is whether WCA provides a distinct architecture primitive that can be controlled, attributed, scaled, and combined with future interfaces.

6 Interfaces: From Fields and Mazes to World Tokens

WCA consumes vectorized state tokens. It does not currently consume text tokens in the language-model sense. A field, maze, or future multimodal state must be mapped into H_t.

6.1 Maze interface

For a maze, the node set $\mathcal{V}$ corresponds to grid cells. Input channels include wall, start, goal, coordinates, and open-cell indicators. The output is a potential or distance-like field used for greedy navigation. The functional metric is not just field MSE; a useful maze field must support path rollout from start to goal.

6.2 PatchMean field interface

For a reaction-diffusion frame

$$X_t \in \mathbb{R}^{C\times H\times W},$$

PatchMean partitions the field into N patches and maps each patch to a token by averaging:

$$z_i = \frac{1}{|\Omega_i|} \sum_{(u,v)\in\Omega_i} X_t(:,u,v).$$

PatchMean is auditable but lossy. It preserves low-frequency patch statistics and discards within-patch structure.

6.3 Learnable patch interfaces

A learnable patch interface replaces the average with an encoder:

$$z_i = g_\eta(X_t|_{\Omega_i}),$$

where g_η may be an MLP stem or a convolutional stem. This can preserve features that PatchMean discards. It also creates an attribution question: does the improvement come from g_η alone, or from WCA's recursive core? This is why the report includes tokenizer-only, outer0, token MLP, and token Conv controls.

7 Training Objectives and Metrics

The field-prediction tasks use supervised prediction. Given target field or token labels Y_{t+h} and prediction $\hat{Y}_{t+h}$, the main loss is mean squared error:

$$\mathcal{L}_{\mathrm{MSE}} = \frac{1}{|\Omega|} \sum_{q\in\Omega} \left(\hat{Y}_{t+h}(q) - Y_{t+h}(q)\right)^2.$$

Relative L^2 error is

$$\mathrm{RelL2} = \frac{\|\hat{Y}_{t+h} - Y_{t+h}\|_2}{\|Y_{t+h}\|_2 + \epsilon}.$$

For token-level WCA experiments, the formal metric is token-space MSE unless a native/full-field bridge is explicitly declared. A patch-repeated visualization can look blocky while still having strong token-level MSE. Conversely, a full-field model can look visually sharper while answering a different metric question unless reduced to the same token contract.

8 Evidence Tree

The project is organized as an evidence tree rather than a sequence of ad hoc experiments.

Stage	Question	Representative evidence
Establish	Does WCA show a real signal?	Maze potential fields, WeatherBench-style transfer, V25 recursion ladder.
Attribute	Does the signal come from WCA core?	Tokenizer-only, outer0, token MLP, token Conv controls.
Scale	Does signal improve with N, D, recursion, data, and compute?	V25 recursion depth; V27 N=256 diagnostic boundary.
Generalize	Does WCA transfer to broader families?	Future PDE families, native rollout, unified physical latent states.

This evidence tree is important because an architecture project can easily confuse implementation improvements with architecture improvements. A better patch encoder, a larger decoder, or a lucky checkpoint can all improve numbers without proving the WCA core. The attribution stage exists to separate these possibilities.

9 Maze Experiments

Maze experiments were the first test of WCA as a world-field architecture. The task asks the model to form a goal-conditioned potential field over a grid. A good field should allow greedy descent from the start cell to reach the goal through a valid and preferably optimal path.

These experiments matter because they test functional structure. MSE alone is insufficient: a low-MSE distance field can still contain a local minimum that traps a greedy path. The useful metrics include path success, path optimality, loop rate, goal rank, local-minima count, monotonic descent accuracy, and neighbor-order accuracy.

The maze branch produced two lessons. First, WCA can form useful potential-field behavior at small scale. Second, harder mazes expose shortcut learning, incorrect attraction basins, local minima, and loops. These failure modes shaped the later protocol: functional metrics and guardrails became non-negotiable.

10 WeatherBench-Style Experiments

WeatherBench-style experiments tested whether WCA could leave synthetic fields and operate on real multivariate Earth fields. The compact branch used variables such as 2m temperature, 10m wind components, and mean sea-level pressure. WCA improved over persistence and approached compact baselines in selected horizons.

The larger 240x121-input branch showed that dense WCA can run in a more realistic field setting and improve over persistence. The strongest raw/full field baselines retained better native visual detail in some settings. This branch is therefore best interpreted as real-field transfer evidence: WCA can operate on real multivariate states, but the current interface is not a final weather model.

11 PDEBench Reaction-Diffusion Experiments

PDEBench reaction-diffusion became the strictest current testbed because it supports controlled horizons, trajectory-safe splits, token-level contracts, matched controls, and rollout diagnostics.

The source data are trajectory-based. A strict split assigns entire trajectories to train, validation, and test sets before constructing temporal windows. This avoids trajectory leakage and prevents windows from crossing trajectory boundaries. The reaction-diffusion field has two channels over a 128×128 grid. The primary horizons are h1, h2, h4, and h8.

11.1 V25 recursion-depth ladder

The V25 recursion-depth ladder tests whether outer recursion depth affects long-horizon behavior. Under the simple PatchMean interface, shallow WCA is weak at short horizons, but deeper recursion improves h8 behavior. This is important because it links performance to the WCA mechanism: repeated world-state synthesis matters for longer-horizon prediction.

This result is establish-stage evidence. It does not by itself prove dominance over all baselines, but it supports the claim that recursion depth is not an irrelevant implementation detail.

11.2 V25b/V25e learnable-interface attribution

PatchMean discards within-patch spatial structure. V25b introduced learnable MLP-stem and Conv-stem interfaces to test whether this bottleneck caused the short-horizon weakness. The improvement was large, especially for the MLP-stem branch.

The next question was attribution. If the encoder and decoder can solve the task alone, then the result is not evidence for WCA. V25e-r3 therefore compared full-core WCA against token MLP, token Conv, tokenizer-only, and outer0 no-recursion controls. The source-audit pattern supports WCA-core contribution at h1, h2, and h8, while h4 includes an explicit outer0 exception. This exception is scientifically useful: it shows that the report is not hiding a horizon where recursion is not the best explanation.

11.3 V25d h8x2 rollout

Direct h8 prediction is not enough to test world-state behavior. V25d evaluates a two-step h8 rollout: predict h8, convert the token output back into the next input representation, and predict another h8 step toward an effective h16 endpoint. The rollout remains token-level; it is not native full-resolution autoregressive simulation.

Table 3: V25d h8x2 token-level rollout main numeric result. Each row uses 64 fixed evaluation starts.

Model	Role	Seed	Samples	Mean MSE
MLP-stem WCA full core	formal token-level model	15101	64	$1.20737 \cdot 10^{-5}$
MLP-stem WCA full core	formal token-level model	15102	64	$2.27541 \cdot 10^{-5}$
Tokenizer-only control	negative control	15201	64	$9.38550 \cdot 10^{-5}$
Tokenizer-only control	negative control	15202	64	$1.38443 \cdot 10^{-4}$
Outer0 no-recursion control	negative control	15201	64	$9.60752 \cdot 10^{-5}$
Outer0 no-recursion control	negative control	15202	64	$1.14411 \cdot 10^{-4}$
FNO raw-field anchor	context only	15001	64	$4.28589 \cdot 10^{-5}$
FNO raw-field anchor	context only	15002	64	$7.71818 \cdot 10^{-5}$
U-Net raw-field anchor	context only	15001	64	$5.12693 \cdot 10^{-5}$
U-Net raw-field anchor	context only	15002	64	$3.87670 \cdot 10^{-5}$

The formal same-family comparison is between WCA and the tokenizer-only or outer0 controls. In both reported seeds, full-core WCA is lower-error. The raw-field FNO and U-Net rows contextualize field-model behavior but are not token-equivalent formal controls unless reduced to the same token contract.

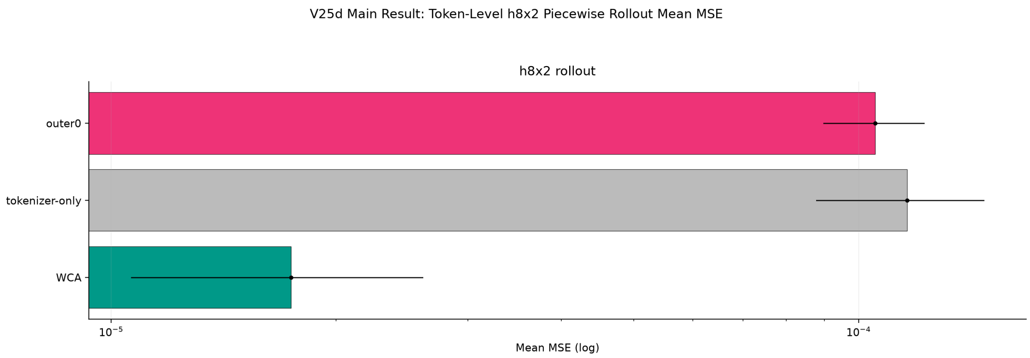

Figure 3: V25d h8x2 rollout mean MSE. The main formal comparison is token-level WCA against same-family tokenizer-only and outer0 controls.

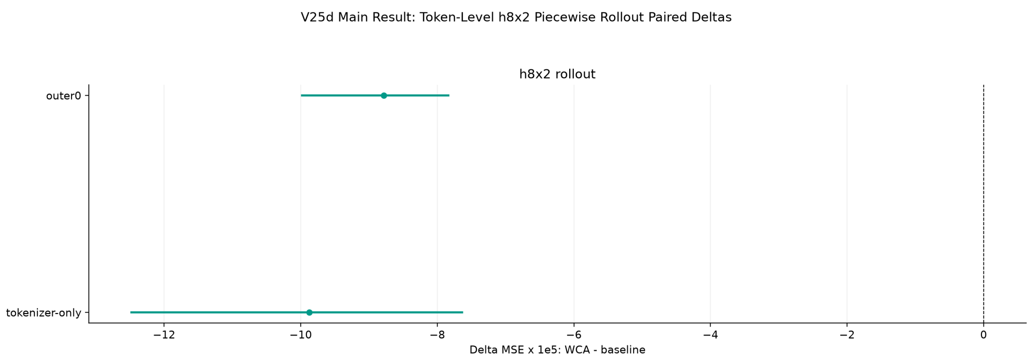

Figure 4: Paired rollout deltas over fixed starts. This supports the h8x2 token-level rollout claim but does not replace larger multi-seed replication.

12 Qualitative Panels and Metric-Space Boundaries

The qualitative figures are useful but must be read correctly. WCA token-level outputs are visualized by repeating predicted patch tokens back into image patches. This makes the field look blocky by construction. Raw/full-field FNO or U-Net outputs can preserve within-patch detail. Therefore qualitative panels show representation boundaries; they do not establish native visual superiority for WCA.

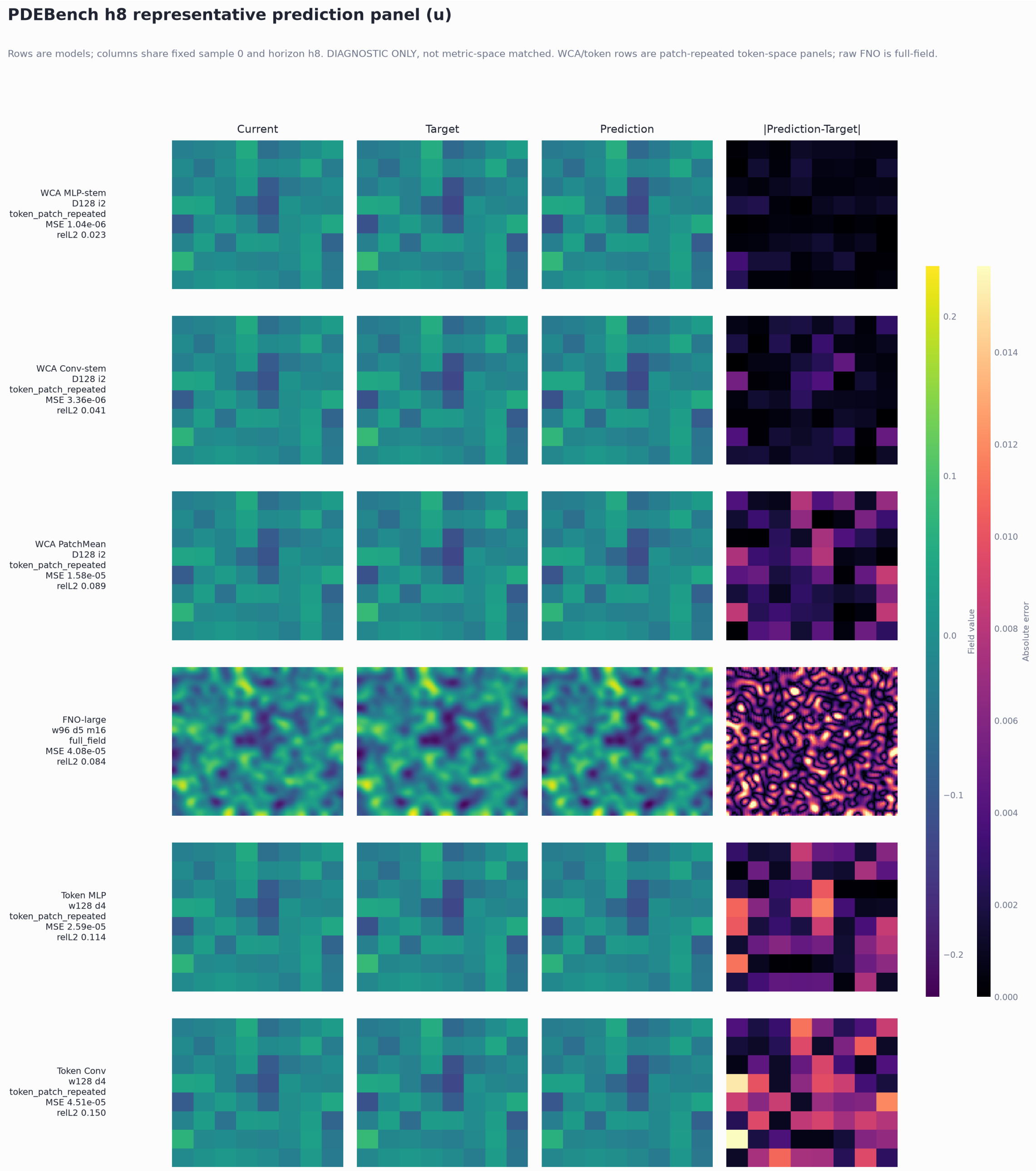

Figure 5: PDEBench h8 qualitative panel for channel u. WCA/token rows are patch-repeated token outputs; raw anchors may be full-field outputs.

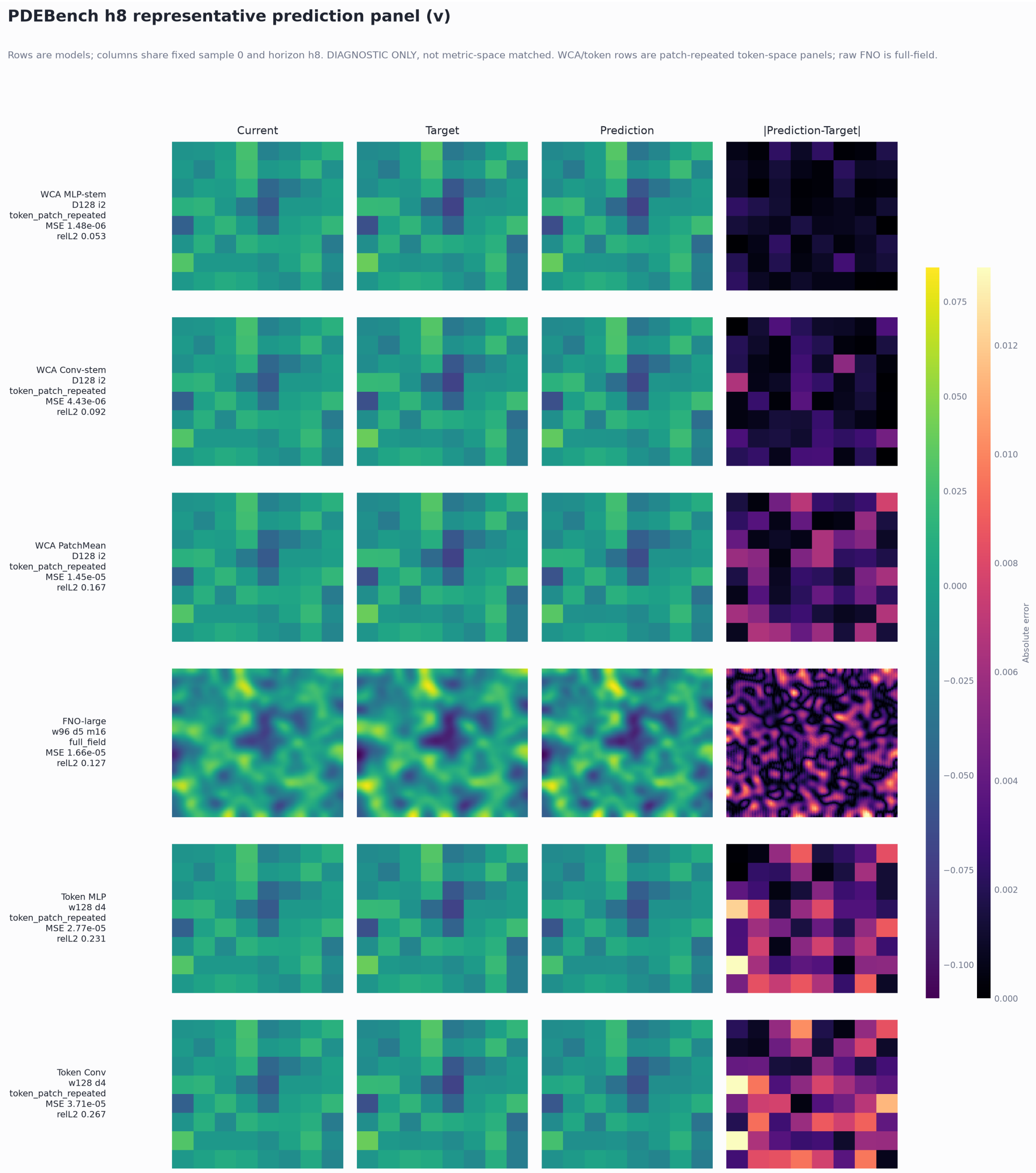

Figure 6: PDEBench h8 qualitative panel for channel v. Quantitative claims must come from explicitly declared metric space.

13 Negative Results and Lessons

A technical report should record failures because they explain the next research program.

13.1 Maze failure modes

Harder maze families exposed local minima, loops, and shortcut learning. This means WCA can form useful fields but still needs topology-aware losses and functional rollout metrics for navigation.

13.2 PatchMean bottleneck

PatchMean is auditable but lossy. Its short-horizon weakness helped identify the need for learnable interfaces. This is a positive scientific outcome: the interface bottleneck became experimentally visible.

13.3 N=256 PatchMean boundary

N=256 PatchMean WCA underperformed matched token baselines. This should not be read as a theorem that WCA cannot scale. It shows that increasing token count under a weak interface and dense implementation can expose capacity and optimization limits. The next scaling test should use learnable interfaces and capacity-safe recursion ladders.

13.4 Decoder and tokenizer confounds

Learnable interfaces improve performance, but they also create attribution risks. A large decoder can become a model in its own right. The correct response is not to avoid learnable interfaces; it is to include tokenizer-only, outer0, decoder-size, and token-equivalent controls.

14 Current Interpretation

The current evidence supports four claims.

1. WCA is a coherent architecture distinct from ordinary GNNs, attention, and direct neural-operator maps.
2. Recursion depth matters in the tested PDEBench reaction-diffusion setting, especially for h8 under PatchMean.
3. Learnable interfaces reveal stronger WCA behavior than PatchMean, and source-audit controls suggest that the WCA core contributes beyond the interface alone on several horizons.
4. V25d h8x2 token-level rollout provides the cleanest current formal evidence that the WCA core can support iterative long-horizon token dynamics.

The current evidence does not finish the research program. The strongest formal rows are token-level. Native full-field decoding, broader PDE families, longer rollout, stronger replication, efficient dense-equivalent kernels, and unified physical latent interfaces remain open.

15 Roadmap

15.1 Native and learnable interfaces

The next major model direction is a native WCA field interface. The model should preserve more within-patch information without allowing the decoder to hide the WCA core. This requires interface ablations, decoder controls, and token-equivalent baselines.

15.2 Broader PDE families

Reaction-diffusion is only one physical family. The next experiments should include advection, Burgers-type dynamics, wave propagation, diffusion variants, and Navier-Stokes-like flows when strict split contracts are available.

15.3 Rollout

h8x2 is a useful diagnostic, but a stronger world-state test needs longer rollout, spectrum or energy-drift metrics, native-resolution error, and stability checks.

15.4 Efficient dense-equivalent implementation

Full dense WCA is expensive. The first systems goal is not to change the mathematics, but to preserve dense semantics while improving memory and GPU throughput. Later sparse, hybrid, or multiscale variants should be introduced as explicit variants with equivalence or ablation gates.

15.5 Unified physical latent

The long-term direction is a unified physical latent interface. Images, tactile grids, simulator states, weather fields, maze states, and other modalities could be encoded into a shared WCA state space. This is a future branch, not a current result, but it is the natural extension of WCA as a state-space architecture.

16 Reproducibility and Release Package

The public release should include:

- this technical report in English and Chinese;
- source LaTeX files and compiled PDFs;
- figures and tables used in the report;
- a claim table separating formal, source-audit, secondary, diagnostic, and future evidence;
- artifact manifests with hashes;
- GitHub, Hugging Face, and Zenodo release notes;
- a reproduction guide that explains the strict split, horizon, token, and checkpoint contracts.

17 Conclusion

World Cellular Automata is a recursive local-world architecture for world-state modeling. Its defining computation expands a global state into center-indexed local worlds, evolves those worlds through pair interactions, and recomposes evolved centers into the next global state. This structure gives WCA a clear identity: it is not a generic MLP stack, not ordinary graph message passing, not transformer attention, and not a direct neural operator.

The experiments recorded here show that WCA is worth continued study. Maze experiments motivate world-field formation and reveal topology failures. WeatherBench-style runs demonstrate real multivariate field applicability. PDEBench reaction-diffusion provides the strongest current evidence: recursion depth matters, learnable interfaces reveal stronger dynamics, and h8x2 token-level rollout supports WCA-core contribution beyond interface-only and no-recursion controls.

The research program is therefore clear. WCA should be developed as a candidate world-state modeling primitive, with stricter attribution, stronger native interfaces, broader physical benchmarks, longer rollout, and efficient dense-equivalent implementation.